

Meeting Minutes

Group 8 - C.A.V.E: Cave Assessment and Visualization Equipment

Sept. 11, 2025

Attendees: Abdul, Tabish, Berk, Andrew (online)

Notetaker: Nicholas

Agenda:

Team Technology

- Messaging/Communication: Teams
- Files- onedrive/sharepoint, eventually all Github (especially source files)
- CI/CD- optional since we're tron, should do some CI as is convenient and helps us, should be fire and forget
- Berk makes the repo
- Refining scope given by Andrew into an email to Dr. Wassying for approval for Berk's entry into the group. Will then slightly rotate it into the group formation message required for the deliverable.
- Regular meeting during tutorial time (first hour), in person
- Andrew as Liasion, chair
- Nicholas as scribe
- Python for now, cython exists if speedup needed
- Can split computation into recording data/hardware interface, processing is separate task on another computer. Want to start over on that code, could pick fresh start. Various C/C++ libraries for these tasks
- Possible stretch goal to have a backend, database to back up info/sync btn pi and server
- Probably have a linter, Berk will see to it. Can follow formal convention, put issue number in commit message: <https://www.conventionalcommits.org/en/v1.0.0/>

Future Work

- Explore existing literature (aka the code from the last group, or Andrew's from last year)
- Figure out how the power delivery board works, determine whether we can reuse
- Sent out GitHub IDs to Andrew to share the code with us

Deliverables

- Next deliverable is September 21st, 10 days from now; form teams, project description.
- Berk has emailed the professors about his case, will move towards the team formation deliverable based on that.

Sept. 18, 2025

Attendees: Abdul, Tabish, Berk, Andrew (all online)

Notetaker: Nicholas

Project Goals and Development Plan

Plan:

- Work asynchronously throughout the week on goals, add requirements and constraints as they relate to goals.
- Work on this whenever have time over the next week
- Set main goal small, add stretch goals for other feasible
- Dev Plan rough outline
- Review existing materials from research project and last year's capstone, going over source code, power pcb, see what is useful for us
- Make contact with old capstone group to understand that power pcb end of Sept
- After, test sensors to see how well they work, and roll into proof of concept deliverable Oct
- For PoC itself, mainly need to demonstrate software can work with data from the sensors, fusing the two. Electrical and mechanical are minor considerations, test sensor→sfwr model
- critical point of POC is ensuring sensors work for our goal
- Mech, elec, sfwr designed/implemented in parallel, feedback between them, iterations
- Main shortcomings to improve this year for rev0: ToF camera/sensor was not really working last year
- Reusing Pi5, could improve electrical system for rev0. Could also want something more efficient for sensor collection, data processed elsewhere. Do PoC and then re-evaluate

Development:

- Andrew plugged in the TOF camera and Windows sees it as a serial device
- Sensor documentation:
<https://wiki.sipeed.com/hardware/en/maixsense/maixsense-a011/maixsense-a010.html>

Deliverables:

Project Goals and Development Plan - Sept 28 (11 days)

Sept. 25, 2025

Attendees: Abdul, Tabish, Berk, Andrew

Notetaker: Nicholas

Project Goals and Development meeting (*deliverable on the 28th*)

- Break down document into chunks, one person is responsible for completing it, two more for reviewing it
- Finish writing sections by Saturday, finish reviewing by Sunday
- Add rough goals to doc together during meeting
- Also hash out the steps involved in project development
- Prioritize setting up Git repo, evaluate using Dr. Smith's template
- Split the work out, everyone has ~2 writing items to complete by Saturday, 3-4 reviewing tasks

Oct. 2, 2025

Attendees: Abdul, Tabish, Berk, *Andrew (Away)*

Notetaker: Nicholas

Agenda:

GitHub

- Used Dr. Smith's template, has a lot extras/fluff that we won't probably use like in docs
- Main thing to keep is the action compiling the latex files, will be useful for future documentation
- All future docs in the Github as tex
- Convert meetings notes and commit
- Same for goals/development plan
- Use issues to track who does what -link commits to issues, assign to people
- CI/CD: some tools to autogenerate issues if there are problems with the code
- Use issues for actual tasks to complete in repo, ie meeting minutes, code additions, documentation
- Add issues for ci/cd, notes, etc.
- May be an issue with permissions, no one can assign issues or create commits

Action Items

- Github permissions from Andrew
- Add sufficient issues to get us to the PoC (Oct 27)
- Complete the setup tasks before the next TA meeting
- Meet with Giamou before PoC, possibly during this meeting time

Oct. 9, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

TA Meeting - 3:45pm - Dr. Wassyng, Vasily Kapustin

Agenda

- Verify PoC deliverables
- Discuss main branch protection
- Talk about the previous years' attempt, what we can take from that

Minutes

- Went over summary of the project with Dr. Wassyng and Dr. Rapoport, cave mapping goal
- Got a hold of one of last years' group. On the topic of reuse, it's up to us how much to reuse. Build on other projects with most projects, just acknowledge.
- Last year didnt finish algos, apparently had trouble with the helmet. It didn't see everything they wanted, was unnatural to use.
- Talk with Giamou a lot to start- this will become difficult
- Easier to write your own algorithms than trying to debug someone else's. Will be easier to start fresh
- ToF sensor works with vendor software, in Thode
- Proof of concept deliverables- not a prototype, just think about what is critical for success. Show some kind of proof that we can overcome the main hurdle
- Quick and dirty, feasibility proof. Does not have to be used in final prototypes/iterations.
- Informal presentation, tell problem (1-2 of them) show why it works. Previous videos to base off off
- Written contract: submitted Friday Oct 31. Things that we are guaranteeing that we will have done. High level goals, no nice-to-haves, underpromise, overdeliver.
- *Berk arrives*
- Written contract is conditions for success; shouldnt be trivial.
- Do a good job on version control. Try and get (code) architecture done early, set interfaces (internal sw, sw/hw).
- Do sfwr early- some groups skimp on unit testing and just do integration. Having the lower level verification eases debugging
- Backups are a good idea, disable force pushes (history rewriting).
- Chat with Vasily, he's very good with software (and toasters). He is not usually at these meetings, more of an advisor. (they have other classes and commitments)
- Have a reasonably regular meeting with him. He'll be vocal about changes we should make.
- *Reviewing* and reusing their electronics counts for the electrical requirement- "debugging is more difficult than writing"
- Dont be as smart as possible when coding, because then you're not smart enough to debug it. Keep it simple. Optimize after it works.

Action Items

- Work on proof of concept over reading week break (due week of Oct 27)
- Work on written demo contract (due Oct 31)
- Talk with Dr. Giamou and our TA Vasily

Supervisor Meeting - 4:10-5:00pm

Agenda

- Plan future meetings
- Identify points of clarification
- Talk with Dr. Giamou about what we should do for the PoC

Minutes

- Before meeting with Giamou, talking about reading week plans
- Will meet online during the regular slot on reading week, to check in
- Simple slideshow with some of last years' stuff, whatever work we have done for the proof of concept.
- electrical distribution board appears to have been open sourced
- PoC: show sensor working with a laptop or Pi, fusion?
- Stick with MIT license
- Review the CI/CD settings so that we don't get 3 emails every time documentation is changed
- Should create a milestone for the PoC and start putting issues under that to distribute work
- Try out decomposing issues, see how the hierarchy works in Github- start big, break into small
- New/different sensors: keep current for PoC, change later if needed to achieve goals or requirements
- Make issues as we think of them, assign them as needed
- Can reconfigure sonarqube to ignore documents file, may disable actions for now
- Tabish can also help with unit tests; more of a later problem. Will know more about what to test once we have more code completed.
- *Meeting with Dr Giamou*
- Updating, telling him about deliverables. Need to figure out what the PoC pain points are.
- Getting a couple of the sensors integrated, have 3 weeks to work on PoC
- What went well last year and what didnt go well?
- Werent able to do point cloud alignment with ToF sensors. Try and get them up and running or decide on different sensor, or choose brighter env.
- We found that ToF works, could work with the vendors software. Need to get it clean enough to do ICP, fuse with IMU.
- IMU as complimentary filter, use only for orientation. Gyro/yaw
- 2D lidar was working well indoor. ToF is supposed to help with low light

- To make point cloud, imu gives orientation, accumulate points from one position. Need ToF to do full 6dof.
- For us, lidar gets larger FoV point cloud that isnt ToF, which have narrower views.
- Start with ToF, get them up and running, ICP with them. DOn't need to fuse IMU yet, use with 2d lidar to build point cloud.
- Pose: orientation+position- 6 axes
- Andrew has academic resources for this form last year, textbook on state estimation. Andrew's literature review on subterranean mapping.
- *Andrew*- send us the resources
- Open loop estimation, trusting odometry from ICP.
- Did a demo with the ToF sensor, looks pretty good. Does well in middling lighting within Giamou's office.
- Consider other battery chemistries- LiFePO4?
- Decide on coverage of the two sensors, what fields of view are important. Side to side? Forward? experiment. No need for stereo, overlapping could interfere though
- Consider button to control when imaging, no need for constant movement
- Existing algos to fuse data, openCV, ICP
- For combining IMU with point cloud, do extended filtering. No loop closures. 2d slice not for space, for mapping
- Next meeting should be right after reading week or when PoC is mostly done, 2-3wks from now, would have to be online once his daughter is born. Meet online in November.
- Giamou is in person M/T/Th
- Mondays are a good day- 1:30-3:30, or 10:30-12:30. 20th or 27th depending on PoC, likely at 2:30pm.

Action Items

- PoC due Oct 30 - see above, mainly work on ToF. IMU later, then 2d lidar.
- Add Dr. Giamou to the GitHub - username mattgiamou

Oct. 16, 2025

Attendees: Abdul, Tabish, Berk, Andrew

Notetaker: Nicholas

Team Reading Week Sync Up

Agenda

- Figure out what we need to do for the electives over the next two weeks.
- Catch up on who has done what

Minutes

- PoC and contract is main deliverables
- Example contracts have been posted, they are short
- Some are more formal, could copy and paste format from examples
- Middle ground, or see what it looks like
- Currently using sensors with companies algorithm, only shows realtime map, no persistence
- Apparently are some sensors from the same company that would work well but are out of stock
- Electrical - Unsure what boards 2/3 looks like, may just be in bin with other stuff. Not sure if theyre still needed
- Figure out purposes of all the boards, write down
- Old groups code has also been added, need to go over their code to understand sensor interfaces
- Invited Dr. Giamou to the repo
- Created a github milestone to organize PoC
- Need fusion from IMU to create persistent map from ToF- should be main goal of PoC
- Ask Andrew to bring built PCBs for Nicholas
- IMU is BNO055
- Leaning towards using Pi4, reduced power requirements and we don't need the power onboard
- Some licensing issues for Sonarqube with C++?
- Cython again an option, or alternative static analysis tools
- Textbooks- will add a page with links/resources for us to use - http://asrl.utias.utoronto.ca/~tdb/bib/barfoot_ser24.pdf
- Go over SLAM, sensor fusion
- Document sensor setup so others can replicate- try docker image

Action Items

- Proof of concept due in 2 weeks
- Draft contract as a part of that
- Abdul will work on sensors - try and get raw data from sensor, analyze somehow into a single frame/capture
- Nicholas will dissect schematics
- Everyone should get familiar with last years' code, algorithms

Oct. 23, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

- PoC update on deliverables
- Talk about work over the past 2 weeks, what needs to be achieved

Minutes

- Poc progress- worked on sensors, finish work this week
- PoC is testing ToF sensors by mapping an environment into a 3d model (single frame)
- whole environment of multiple frames is too far for now
- Previous group failed at above stage
- Simple, do on laptop, take look at algorithms
- Previous group has well designed boards, reusable
- They had an issue with knowing where they were - IMU/pointcloud fuse
- They mostly used lidar, which on its own does not really give position
- Taking two pointclouds and determining where we moved would be a great level
- Talk to TA to get back comments on PoC
- Walk in and show it next week, can do slides if we want. Literal demo
- Will improv with feedback to show more things if wanted
- Demo of ToF sensor
- For PoC presentation, brief intro to project, why we chose that, whether it worked or didnt work
- ROS running on Abduls laptop
- Showing depth captured on the sensor, 3m range. Works for our distances, close objects can block some further things
- Robustness important factor for project, demo doesnt instantly break
- Reflections from water could affect performance -hazard
- Hazards will be both harm, and potential inhibitions to project- "loss"
- Depth demo on ROS working, gets 5 frames per second. Plan out how this will be shown for PoC, highlight what were worrying about that we can do
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Action Items

- Complete work on PoC, informal slides or talking order to organize how we present
- Redundant setup for demo if someones laptop doesnt work
- Setup this afternoons meeting

Team Meeting - 4:30pm

Agenda

- Determine remaining work to be done before next weeks PoC deliverable
- Determine whether we will meet with Dr. Giamou next Mondays
- Discuss moving this meeting slot to the (previously lecture) time before the prof meeting

Minutes

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Action Items

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Oct. 27, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 2:30pm

Agenda

- Assign work to be able to present POC and demo contract
- Set up ROS for PoC sensor demonstration
- Move regular meeting time?

Minutes

- Abdul working on C file to extract data from sensor for PoC - to create branch
- Rest working on setting up ROS for redundancy
- Tabish working on one-page pamphlet-y thing to organize info for presentation
- Demo contract- derived from project goals, take bare minimum and rework more formal
- Should be a LaTeX document, formal and formatted
- Appears to be very short, based on sample docs
- ICP would be nice for the proof of concept- could get 2 captures and work on it on the side
- Going forward, weekly meeting is moved from Thursday at 4:30pm to Thursday at 10:30am, immediately before the meeting with the profs
- Giamou working from home for next 4 weeks
- Installing ROS is a pain, does work best on Ubuntu but very picky about its compatibility
- Contract should have the main goals, with the direct goal being a low-light room rather than a cave
- Synthesize that with the examples, at a medium level of legalese
- Berk and Tabish have ROS running, will test ToF sensor on their systems

Action Items

- Abdul to take lead on C program for capturing and exporting a point cloud- doable today
- Tabish to create PoC presentation media
- If he has time, Andrew will try ICP on two point clouds
- Nicholas to do the demo contract
- Berk and Nicholas to both have ROS as backup

Nov. 06, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 10:30am

Agenda

- Share progress
- Break down next deliverable- Rev0 requirements and hazards

Minutes

- Next deliverable is requirements and hazards, due Nov 16, worth 8%
- Lectures suggested FTA for hazard analysis
- Simple ICP algo, whats next fro the code?
- next step is continuous running, adding new point clouds
- can change to point-to-plane ICP, more accurate and reliable. Better for 3d shapes
- Need to add normals to points for that, have libraries
- Take transformation matrix from each point cloud mapping to generate an object
- Fusing 2 TOF sensors is another challenge. Have transform between two posns. Find optimal posns for both
- Prioritize one ToF for now, get second for Rev0
- Berk takes IMU, work to find library, ROS setup. Get reliable data
- Tabish working with ToF input to database
- Still have another meeting and 2 weekends until deliverable, will work async on document, hazard brainstorming
- Scoping out future deadlines, figuring out requirements vs goals
- Continue making issues on github for tasks
- Adding main branch protection
- Use FTA for hazards, easiest. Use existing hazards file from repo template

Action Items

- Nicholas to finish analyzing the Pi hat
- Berk working with IMU
- Andrew, Abdul investigating advancing ICP algos

Nov. 20, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 4:30pm

Agenda

- Talk about next deliverable, what we need to work on
- Future development work

Minutes

- Abdul is working on ICP, made it point-plane
- Need to focus on ICP
- Can add initial guesses to ICP from orientation data to help with this
- Abdul to add documentation for his current ICP code for others to work on it
- Use ROS packages for IMU, should automatically get values
- ROS can poll them at the same time, relate IMU and TOF data temporally
- Attempt to meet after exams online to line everything up for the design deliverable- timezones will be a consideration
- Next deliverable is a large amount of work, huge amount of design, module interfaces and implementation
- Software doesn't need to be completely formal, but whatever communicates the structures needed
- Splitting up code- need to plan out who works on what. Can maybe save lidar for rev1
- Transformation from IMU, TOF, lidar is there to add extra data points
- Closed vs open loop scanning - closing is coming back to where you started, fixing previous scans backwards
- Displaying remaining time could be annoying- 7 segment, or read out binary LEDs
- we have money to spend if needed
- Battery pack could be challenging to get the time remaining.
- Storage space is easier to keep track of
- Andrew has a programmable watch that could connect via bluetooth for control/info
- Would be an easier controller than having someone else press buttons on the helmet
- could design better UI on the helmet to make info easier to get

Action Items

- Ask professor about when the last meeting is

Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

- Ask when last meeting
- Any design document questions

Minutes

- Last meeting will be thursday dec 4, so 2 more meetings
- If we dont need to talk to them, we don't have to, let them know in advance, dont do it all the time
- Sfwreng has a bit more experience with how this documentation should go
- Variables and constants are similar aspects from requirements- difference is reqs are very high level, low detail
- Taking variables into design, process them, answer the how of what happens
- Note which part of the design implements what requirement, or part of a requirement
- Helps keep everyone on same page with how the project should come together
- Also helps with debugging, can identify which requirements are not being followed
- Class will be responsible for some part of reqs, inside will have state variables- dataflow then looks much different from reqs
- MIS is most important, practically end sup as class interfaces. How they talk to each other, the crucial part of sfwr design
- MID can be pseudocode, function tables, not all the way to language syntax to get behaviour
- Something is going wrong if the coding isnt simple after the design
- Req is what (and sometimes, hopefully, why), design is how
- Design will look different from reqs as reality modifies the dataflow
- Can bounce ideas off TAs, Dr Wassyng. Meet whenever needed if a problem is encountered
- C# is Wassyng's recommendation, has typing but some protections for the coders
- Does it work for realtime? Can turn off garbage collector to make it faster
- Polyspace is a good static analyzer from Mathworks - model driven?
- Could look at some model driven tools (simulink style)
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Action Items

- Begin design documentation
- Work one hardware platform, how to be put together

Dec. 04, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 10:30pm

Agenda

- Discuss new items to ask Dr. Wassyng
- Discuss working on the deliverable over the month

Minutes

- Discussing using the budget to get more IMUs to share amongst the group
- Want to ask prof about electrical and mechanical design documentation- are there formats to follow, or use generally the same structures
- Report seems to be expected to be long when looking at past examples
- New IMU runs about \$20, likely worth it to have more hardware flexibility in the group
- Doc includes schematics and interfaces of some of the electrical, platform,
- Our equivalent might be published schematics, CAD models of each sensor, the RPi, etc
- Dimensional drawings exist for sipeed TOF sensor
- We also have schematics of the past group's boards, may include
- Andrew here tues/wed/fri, nicholas also definitely here those days
- Berk 6-8, 13, 15, 18
- Overleaf is a possible option to work on the doc together
- Easiest to use a word document and then translate to .tex, can try out Overleaf
- Dont need to submit pdf source code, just the pdf
- Free plan on Overleaf only allows one collaborator, might as well just use word
- Overleaf is open source, can self host if we really want to
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Action Items

- Ask Dr Wassyng about mech/elec decompose
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Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

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Minutes

- Electrical normally easier, just schematics, io planning or charts
- Mech depends on what we do, cad files etc can do drawings
- Dont want to do too much extra beyond design, jsut including schematics, drawings, with some explanation is good
- Mechanical - can take pictures, show where sensors are, why they are there
- Stick last years' in an appendix, include reasoning about whether we do or dont use it
- Include rationale for the design, going to want to know what engineering decisions have been made
- Do stuff tht helps put it together, not just random filler for the report
- If theres a section we dont think applies, can omit it and give an explanation why we dont think it is needed
- Iterate- prob need to work on sfwr a bit to figure out the structure of the design
- Document top down as if it was perfect from the start
- Do all sfwr design thinking about how to layout the classes, module/class guide with interfaces, interactions
- Do as much design as possible in terms of pseudocode, or function tables, should make code trivial
- Have equation for the algorithms in design document, just have eqns rather than talking about them
- For outside, open-source code, acknowledge where its from, explain why we think its good to use, verify that it works
- Just verify license, we are currently on MIT, if we use GPL we'd have to inherit that
- We can ask Dr Wassyng over the break, he may be around if we need help
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Action Items

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Dec. 18, 2025

C.A.V.E Attendees: Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 10:30pm

Agenda

- Final meeting before System Design deliverable is due

Minutes

- Discussion on sections of the document, who will be where over the holiday
- Splitting up document between us now that we have a better view of its formal
- Nicholas will post MIS MID examples in the chat
- We are aiming to complete the deliverable by its due date or earlier, no sense in dragging it out.
- Possibly call again part way through working on Saturday or so

Action Items

- Nich will post MIS/MID examples
- Everyone has put their names onto sections and will complete that- software subsystems

Jan. 08, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Team Meeting - 10:30pm

Agenda

- Rev0 deliverable

Minutes

- Sensor fusion is main problem
- Need to mount things to helmet to get a proper test run of data to do fusion on
- Reviewing what components we need to build out all the PCBs, mostly the two small boards
- Ros2 has been tried on the Pi (to use the IMU)
- Add more issues to cover all the things we need to complete these
- Did a when2meet to figure out times we're all available this term. Good availability in afternoons early in the week
- Planning a meeting with Dr Giamou this tuesday, and then recurring work periods on tuesday or thursday
- Abdul is the best at CAD out of us, he can likely do the helmet mounts
- Planning out sensor placements- TOF similar to antennae facing outwards at each temple
- LiDAR at the front of the helmet, with the plane crossing vertically in front- bottom of lidar mounted the front of helmet
- Had basically a gopro mount on the helmet, 1/4' thread or smth attached with glue
- No drilling into helmet, double sided tape may work
- Final product we show should be a 3d map from a capture taken several days earlier, time to figure out merging
- Could use test jig to get capture from all the sensors together, even without helmet, to get sensor data to work onto
- Timing component could be a challenge
- Aim for the final product for rev0, but not necessarily the most accurate output
-

Action Items

- Ask about sensor placement at meeting
- Also ask about presentation standards- similar to PoC slides
- Meet on Tuesday with Giamou, bring in helmet to begin mounting
-

Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

- See action items above

Minutes

- Presentation will be more formal, want slides, have the projector
- Same teams setup for projecting the slides.
- Bit more rehearsal, do some practice so we know who's saying what, dont be reliant on the slides
- Should be fluent and concise, dont want to waste time- time will disappear quickly
- no rule on everyone having to talk, cna have others doing other things. work out transitions between presenters
- This will be a rehearsal for rev1 and final
- present 'as if' it were done, afterwards say what we're actually completed
- 12 minutes for the presenation, 3 for questions
- 100% honesty on questions, pass around to whoever the specialist is
- Rev1 not very different if rev0 is good- improve based on their feedback, but mostly on our internal assessments of whats good
- No grade for completion in rev0, small grade for it in rev1- with respect to the contract (which can still be updated)
- Be careful with colours on slides, the projector will wash them out- especially white text on a light background, aim for contrast
- Make it more interesting than bullet points, there are better ways to present info- many pictures, add labels, talk around that
- Sensor placement: link what we're doing with requirements, performance or ergonomic requirements
- Tradeoffs between these requirements
- Talk to Vasily since he has been doing the same sort of fusion on model cars
- !!Remind him that we're interested in sensor fusion, send him something on Teams- used matlab/simulink for it
- If requirements dont necessitate it, we can ignore the lidar- if its not helping, its harming
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Action Items

- Ask him about sensor fusion on Teams

Jan. 13, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Meeting with Dr. Giamou to plot Rev0

Agenda

- Rev0 deliverable

Minutes

- Supports unicorn or front position of the lidar on the helmet
- Need to avoid it sensing the helmet, unicorn avoids a lot of that
- Has the stencils for the PCBs- I(Nicholas) can use that tomorrow
- Mounting tabs- we can 3d print direct mount to the flatter pieces with tape
- Have some docker fixes, should be able to do 1-2 commands to get image working with Ros2
- Test ROS sync of both TOFs, can assume caver may not move between fast enough captures (25hz?)
- Gaussian processes- geostatistical analysis thing- assume signal is gaussianly related through time? smoothes clock jitter
- Have makerspace and solar car resources for space
- If we need a new pi, can order on the capstone dime
- ICP rundown- use both TOF, treat as single super sensor, run at 5hz, little overlap is fine, may be useful for calibration
- Batch offline state estimation- run algo at home
- IMU is ego motion frame, 'centre' gives accel and angular velocity
- motion/odometry from IMU, TOF, use laser to map onto space
- IMU as estimate for ICP, short time period integrated accel- open loop method
- Or, use as complementary filter (double accel integration has high error) assume accel is dominated by gravity, use angular velocity for orientation, roll and pitch without yaw
- May be existing ROS packages to do this, GTSAM is a python/cpp package for it
- Solves opt problem of going between time steps from ICP Ti frames and complementary filters from IMU
- Perception in the Dark paper from 2020 uses the first method we considered
- Use up our budget first, and then Giamou can spot us for mmore if needed
- May meet on next Tuesday, depends where our work is, can meet at the same time every Tuesday if needed
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Action Items

- Use stencils to create some more boards
- Fix names of meeting minutes to make my life easier
- Read over paper
- Test out mounting and capture
-

Jan. 15, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Online meeting due to inclement weather, meeting with Dr. Wassyng cancelled

Agenda

- Rev0 deliverable

Minutes

- Got docker running on a 2gig raspi, that should be enough for our purposes
- Nich checking for bridged contacts, etc on pi hat, finishing through hole
- New IMU uses different software packages from the old one, could have a ROS package
- Talk of creating another docker container for the local processing, depends on process
- Could keep files on rpi and do processing there in interim
- Dataset is the biggest need
- Combining the two TOFs as one larger point cloud is the easiest way to combine those
- Or two processes with each recording one
- Sensors produce data at their own pace, so synchronizing is not very possible without firmware hacking
- Tabish has started on the slides, going to work more on them today
- Abdul scanned helmet, have 3d model, going to work from there
- Model helps get transformations from Sensors
- We're very free next week
- Monday start putting together the helmet, preparing presentation
- Work on ICP over weekend
- Excel file posted on Teams chat for reimbursements, one of us will submit before March 27
- Expensing likely just the power bank and new IMU
-

Action Items

- Andrews goal is capture with 1 TOF and IMU for processing
- Tabish finish the slides
- Abdul doing 3d
- Berk try sensor fusion- IMU, TOF
- Nicholas finishing boards and soldering, get 20V powerbank
- All try to do thode makerspace - has an in-person sign up needed

Jan. 20, 2025

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas, Dr. Giamou

Notetaker: Nicholas

Meeting with Dr. Giamou

Agenda

- Rev0 deliverable

Minutes

- Challenges getting software to work with IMU
- switched to ICP with just TOFs, some drift over longer captures
- gets transformation matrices, multiply together for final
- Forms some shapes using only one TOF
- Not expecting high fidelity with these Sensors
- Want to measure a loop to see that drift is minimal, loop closure in SLAM
- Have a good point cloud created from slow capture of desk
- Got ros1 version from paper running, need to convert bags from ros2 to ros1 formats
- Berk outputs a spreadsheet of pose estimation, couldnt get imu and TOF working together on Pi-versionitis
- position estimation by integration, next challenge is putting them together
- IMU needs to be calibrated- preexisting calib from mfg
- Longer term- for dense point clouds, techniques to assume envs are smooth, optimize a score (Renyi entropy) to find corrections that make the point clouds tighter from poses
- Expensive but works for offline processing like ours
- rev1 consideration mainly
- Versionitis on the pi with ubuntu 24
- Using imu to estimate pose before ICP may not be needed
- Complementary filter recommended to estimate angular velocities for state estimate
- TOF time sync, plus IMU: can maybe sample IMU fast enough to mostly align with TOF, TICSyn possible method for the 2 TOF
- TICSyn guesses the difference in time between sensor captures- offset between clocks/events from each sensor
- or use earlier TOF capt and IMU data, then forward interpolate with later TOF capt to improve estimation
- Mechanical stuff- have TOF holder printed, need some inserts and screws
- Good elastic strength on the clip mechanism
-

Action Items

- Work on capturing larger spaces with the single sensor
- Attempt to fuse IMU with TOF
- Plan for presentation

Feb. 05, 2026

C.A.V.E Attendees: Abdul, Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Pre V&V meeting

Agenda

- V&V documentation discussion

Minutes

- Need to define validation versus verification with Wassyng
- Aim to complete build in february, basically finish stuff well before rev1 docs
-

Action Items

- Ask Dr. Wassyng about how exactly to do validation, what specifics are needed
- Check verification needs, tests for reqs
-

Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

- See action items

Minutes

- V nV difference:
- Validation- are the requirements correct. often discussion, meeting with experts, opinions, or simulation
- would want to see the models that prove the validation
- Harder to find tools that prove this than verification
- Testing is most often verif, but you can do tests that are validation, often domain experts checking behaviour
- Validation is testing that the numbers in the requirement are good enough
- Will be feedback to and updating the requirements
- 3D model simulation to test algorithms as a ground truth- looking at the point cloud and recognizing objects is most important thing
- Have antoher person look over the simulated or real capture, make someone else look at the point cloud and see if they recognize it without having the input- that would be a good validation, validates that the process works
- be honest, record how successful we were in recognizing the objects
- ****Forget everything, do that experiment, but dont split it into validation and verification.

- Try to avoid confirmation bias
- Validating buttons- take photo, show its connected, how to test that it is functional
- intrusive vs nonintrusive testing- hope that the tests dont affect it
- Can try intrusive tests- loss of power, etc.
- Can insert failures in the code manually
- data loss on power failure as an example
-

Action Items

-

Feb. 12, 2026

C.A.V.E Attendees: Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Post VnV meeting

Agenda

- Figure out next steps
- Rev1 docs March 5

Minutes

- Presentation three weeks from today for Rev1
- Andrew to spend a few days during reading week on it
- Should basically be done at that point
- ToF is dying after 1 minute on USB, will attempt to use UART to avoid linux usb drivers
- Check that gpio header is same between pi4 and pi5, as board is designed for 5
- System integration- python or bash script to handle gpio, button turns on scanning
- Systemd to launch script at startup
- Python GPIO libraries work on pi4, not pi5
- Header is fully backward compatible
- Tabish will be away close to two weeks starting today
- For rev1, want integration of buttons to start recording,
- all electrical stuff is basically done, unless we use uart for the Sensors
- Figure out TOF running for longer
- May want a new board to do 2x UART, I2C, buttons/leds
- need to test interference between Sensors
- if sensors are plugged in at same time, scans end up in phase and interfere with each other
- At 180, no interference
- even at 90 degrees apart, reflections cause interference
-

Action Items

- Ask Dr. Wassyng something
- Finalize sensor positions
- Fix 1 minute crash of TOFs
- Figure out manual startup control of sensors to mitigate interference?

Prof Meeting - 11:15am - Dr. Wassyng, Dr. Rapoport

Agenda

- See action items

Minutes

- Dont start all over again for next rev
- 2 tof fusion- may just be better off with 1 than dealing with fusion
- can have the second as a redundant spare
- even at a cant between them, reflections are an issue
- can discuss why we ended up with just one, why it was better
- SOLving the problem would take a long time, may not be solveable with these sensors
- For the persistence, could just reboot very quickly rather than have to stay online for 20min
- Write up how more expensive sensor would have made the project easier, or more reliable. DONT go trying to solve now, make what we have as robust as possible, and discuss the alternatives that could have worked
- COMmon issue in capstone, money constraint holds accuracy back- engineering decision
- Get on with the rest of it rather than trying to die on the hill of the 2 sensors
- Next presentation, address any comments from TA, add some revision history
- Make notes of anything you didnt address from TA feedback
- Document all the changes in reqs, design, etc and talk through
- For rev1 docs, just revise with feedback. Not making all new docs, just have revision history and reason for change
- Focus on building and testing at this stage
-

Action Items

-

Feb. 18, 2026

C.A.V.E Attendees: Tabish, Berk, Andrew, Nicholas

Notetaker: Nicholas

Online Reading Week meeting

Agenda

- Final project push

Minutes

- UART interface ordered for TOF sensors, hopefully more reliable but same ROS connection
- testing IMU integration with TOF, especially longer captures
- Tabish to look into scripts to control hardware interactions
- J-ICP work from Abdul improves the mapping
- Added some cli interactions, timer on the ICP run, switched to new algo
- also calculates normals
- added mesh feature for 3d point cloud to look better (more solid)
- andrew starting work on GTSAM optimized merging of IMU/TOF
- It uses accel and ICP results together
- Soft power off button would be more graceful, improve data integrity
- Use one light to show when storage is over 90% capacity- script to check disk usage (df)
- lidar- design a mount, may be simple to include and worth adding
-

Action Items

- Berk working on testing longer captures, IMU
- Tabish working on hardware-software interaction, script to launch software, control leds
- Abdul continuing probe in j-ICP
- Andrew trying out GTSAM fusion
- Berk to make some more TOF/IMU captures for Andrew+Abdul
- Capture length needs to improve
-

Feb. 24, 2026

C.A.V.E Attendees: Abdul, Andrew, Berk, Nicholas

Notetaker: Nicholas

Meeting with Dr. Giamou

Agenda

- Rev1 push, algo update

Minutes

- Getting loop closure and using GTSAM on just IMU data
- Working on integrating IMU data
- Trying to correct drift
- Using generalized ICP is much slower, got threading enabled, if we need extra oomf can use a CAS server
- Covariance of the ICP result- can we get this? this helps GTSAM (by default give it identity)
- Hallway problem elongates the covariance, harder to tell in one direction
- Once we have accurate factor graph/poses, can add in lidar slices fairly easily
- Nontrivial to get covariance- will probably leave for now
- UART hasnt been tried yet, hopefully easy in ROS and hopefully fixes dropouts
- Double integrating accelerometer difficult, later step. Just use IMU for orientation.
- GTSAM has class for IMU, may have a built in tool to help us
- May be able to manually indicate loop closure points, possible off the shelf solution for 3d point clouds- ours is noisy and small so may be difficult
- Open loop as a first step is good, closure will be difficult with our field of view
- Fine to manually add loop closure is fine, plus researcher in control means they can go in a loop actively
- Helmet mounts mostly done
- Getting transformation matrices between sensor placements may be a problem.
- may retry double integration with bno85
- Getting dropouts between i2c imu and pi
- check in before caving- they are dangerous. Giamou knows what hes doing
- Expo april 7, may want to cave end of march. Giamou to start on forms now for faculty/uni approval

Action Items

- Dial in algorithms
- Physical integration
- Ask Dr. Wassyng on Thursday if its the same procedure for a field trip vs Andrews summer research- if not have him email

Mar. 5, 2026

C.A.V.E Attendees: Abdul (virtual), Tabish, Andrew, Nicholas

Notetaker: Nicholas

Meeting with Dr. Wassyng, Dr. Rapoport

Agenda

- Rev1 final blockers

Minutes

- abdul calling in, berk couldnt make it
- Cave forms- can we use the same as from the summer research project
- Nothing special in the forms, as long as hes doing forms thats fine
- UART fix- loss of FPS- can talk about what we expected, what we actually got, how we would improve it
- we've added loop closure, and that library has IMU support so could plug in better. Still manual closure
- IMU was a problem for two projects last year, and is our current biggest challenge
- Minnick could also be a good person to ask about IMU
- Sensor itself has two paths, could possibly use more raw sensor data
- Choose smth and see what you can do with it,
- We know the theory but the practice in code is a sticking point
- Ask Dr Minnick and see if he has hints or leads
- Documentation all still due at the same time, no shift due to presentation
- Do documentation where it helps us, don't spent hours in it when we could be getting the project to work
- Say why we havent implemented any feedback that we choose not to incorporate into the documentation
- Test case of having one person generate and another try to recognize it is still very useful case that we should be trying to do
- Biggest worry is getting a recognizable cloud after the fusion
- Good research results are not just success, but info that you cant do it this way is a reasonable outcome

Action Items

- Contact Dr. Minnick about IMU options, see if there's a magic bullet
- System integration
- 'Open loop' verification- have someone unfamiliar with the scene look at the point cloud and try to identify features

Mar. 17, 2026

C.A.V.E Attendees: Abdul, Andrew, Nicholas, Dr. Giamou

Notetaker: Nicholas

Meeting with Dr. Giamou

Agenda

- Post Rev1 catch up and caving information

Minutes

- getting forms together may not come in time for approval
- conditions may not be favourable for caving on march 23 given the cooling, will have a melt
- SHowing Giamou slides from the rev1 presentation to update on project status
- issues with recording, fusion
- TOFs have low resolution, poor connection stability (likely due to cost)
- Worried about being able to collect cave data
- Have GTSAM working with IMU data, can use either rotations or add a preintegration with velocity, but with kinda of poor results
- Manual loop closure still
- Filtering is iffy sometimes, removes excessive amounts of data
- Pyramidal effect around cone of each point capture
- Once we have cave data, may need to add more manual intervention to tell ICP what results are good or bad
- Giamou agrees on most of demo contract, 1.1 most likely to be difficult. 1.3 remains to be seen
- For a capstone demo, we can do a small room, show that cave worked or didnt work to a certain amount
- Experimental/exploratory, learning from the design and we can present these findings
- Action items:
- March 27 poster deadline
- can add stuff on top of it later, or show on the laptop during expo
- reminded that we cant go to caves alone (without Giamou)
- want to integrate everything on the helmet on thursday (mar 19)
- At this point, more human intervention is needed, or tuning uncertainties/improving preintegration
- Due to offline processing, can randomly initialize. Currently initializing poses to identity, could use open loop ICP estimate
- More sophisticated would be to form optztn problem ass least squares, drop rotation matrices, solve linear system

- sensor quality likely limiting factor at this point
- Covariance matrix for ICP currently random, theres a paper defining it but we could do better-datasheet may provide values for pointcloud
- sensitivity analysis method that adds noise and compares to original ICP result

Action Items

- Final integration and testing in a basic environment
- Finish documentation, start on poster
- Go caving